Python Basics for Cybersecurity

Day 1: Introduction to Python and Cybersecurity

- What is Python?
- Uses of Python in cybersecurity
- Ethical hacking and legal permission
- Course tools and rules

Day 2: Python Installation and First Program

- Installing Python and VS Code
- Creating a .py file
- Using print()
- Writing comments

Day 3: Variables and Data Types

- Creating variables
- int, float, str, and bool
- Variable naming rules
- Using type()

Day 4: Input and Output

- Taking input using input()
- Displaying output using print()
- Type conversion using int() and str()
- Using f-strings

Day 5: Operators

- Arithmetic operators
- Comparison operators
- Logical operators
- Simple expressions

Day 6: Strings

- Creating and joining strings
- String indexing and slicing
- lower(), upper(), and strip()
- String length using len()

Day 7: Conditional Statements

- if, elif, and else
- Indentation in Python
- Comparison conditions
- Input validation

Day 8: Loops

- for loop
- while loop
- range() function
- break and continue

Day 9: Password Strength Checker

- Password-length checking
- Uppercase, lowercase, and digit checking
- Special-character checking
- Password-strength result

Day 10: Lists and Tuples

- Creating lists and tuples
- Indexing and slicing
- Adding and removing list items
- Looping through lists

Day 11: Dictionaries and Sets

- Key-value pairs
- Adding and updating dictionary values
- Looping through dictionaries
- Using sets to remove duplicates

Day 12: Functions

- Creating functions using def
- Parameters and arguments
- Return values
- Reusing functions

Day 13: Lab Asset Information Tool

- Collecting device details
- Storing details in dictionaries
- Displaying asset information
- Saving basic asset records

Day 14: Reading Files

- Opening files in read mode
- Using read() and readlines()
- Reading files line by line
- Using with open()

Day 15: Writing Files

- Write mode and append mode
- Using write()
- Adding new lines
- Creating text reports

Day 16: Exception Handling

- Understanding runtime errors
- try and except
- Handling invalid input
- Handling missing files

Day 17: Simple Report Generator

- Taking user details
- Formatting report content
- Saving reports to files
- Testing report output

Day 18: Networking Basics

- IP address and hostname
- Ports and common services
- TCP and HTTP basics
- Client-server model

Day 19: Python Socket Module

- Importing socket
- Getting hostname and local IP
- Creating a TCP socket
- Setting timeout and closing socket

Day 20: Local Service Connection Check

- Understanding localhost
- Using connect() and connect_ex()
- Checking one approved local service
- Handling connection errors

Day 21: HTTP and Requests Library

- URLs and HTTP requests
- GET request using requests
- HTTP status codes
- Request timeout

Day 22: Website Information Collector

- Taking a URL as input
- Getting status code and headers
- Handling request errors
- Saving website details

Day 23: Logs and Regular Expressions

- What are system and web logs?
- Reading a sample log file
- Introduction to the re module
- Finding IP addresses

Day 24: Simple Log Analysis

- Counting IP addresses
- Finding repeated activity
- Counting error codes
- Creating an analysis report

Day 25: Final Project

- Selecting a project topic
- Writing and testing code
- Preparing project output
- Demonstrating the project